2.3 EDA Project Setup and Hypothesis

- *"I'm starting an EDA project. Can you create a minimal project directory structure?"*
- *"Here is the business context and objective I am working on: <paste context>. Task: Create a Problem Definition Document that clearly outlines: (1) business context, (2) project objective, (3) three KPIs, (4) three key hypotheses. Save this document as a text file."*
- *"Could you explain the KPIs in simpler terms?"*

2.4 Data Understanding and Profiling Techniques

- *"I have collected the datasets and placed them in the 'raw' folder under 'data'. Please review them and first create a data dictionary as a text file. Then, evaluate whether the available datasets are sufficient to test the defined hypotheses. If there are any gaps, please document them in the same text file."*
- *"Do these gaps hinder the analysis, or can I still perform a meaningful analysis with the current data?"*

2.5 Constructing the Analytical Dataset

- *"Next, I would like to perform descriptive analysis. For this, please engineer the necessary features and, if required, combine the datasets to prepare a dataset that can also be used for future hypothesis testing."*

2.6 Data Cleaning and Domain Alignment

- *"On the processed dataset, please perform a thorough check for missing values, inconsistencies, and other data quality issues. Then, provide a clear plan to address these issues. Once I review the plan, you can proceed with the cleaning."*
- *"Approved"*

2.7 Introduction to Univariate Analysis

- *"Okay. Now, please begin with univariate analysis. Visualize the relevant variables and add the plots to the notebook so I can review them there. Before that, create a notebook named '01_univariate_analysis.ipynb'"*
- *"@01_univariate_analysis.ipynb What is the distribution of retention across different customer cohorts (e.g., first-month users vs later-month users)?"*

- *"@01_univariate_analysis.ipynb What is the frequency distribution of each demographic variable individually (city, age groups, gender, device)?"*

2.8 Introduction to Bivariate Analysis

- *"Okay, let's start with bivariate analysis. Can you help me visualize the necessary plots for the hypotheses I am going to test? Strictly, don't perform hypothesis testing. Before that, create a notebook named '02_bivariate_analysis.ipynb'"*
- *"@02_bivariate_analysis.ipynb How does retention differ across delivery experience levels (delivery delay, on-time rate, and CSAT)?"*
- *"@02_bivariate_analysis.ipynb How does retention vary with customer spending behavior (average order value and items per order)?"*

2.9 Introduction to Multivariate Analysis

- *"Okay. Let's start with multivariate analysis. Can you help me visualize the necessary plots for the hypotheses I am going to test on? Strictly, don't perform hypothesis testing. Before that, create a notebook named '03_multivariate_analysis.ipynb'"*
- *"Now that we have completed the univariate, bivariate, and multivariate analyses, please document the key observations in a separate file, structured as a narrative that presents the findings as a cohesive story."*

2.10 Introduction to Hypothesis Testing

- *"Okay. Now, let's begin the actual hypothesis testing. Please review the problem definition file and create a new file named 'Hypothesis Testing.' In this file, for each hypothesis, include: (1) the statistical test to be performed, (2) the subset of data required, (3) any other necessary details (assumptions, metrics, etc.). First, create the file. Once I review it and give approval, you can proceed with the analysis."*
- *"It looks good, but I have one suggestion: please test only the first two hypotheses and leave out the last one."*

2.11 Synthesis of Insights and Report Development

- *"Can you create a well-structured report from all the analysis and present it as an HTML page? The report should include the problem statement, data overview, analysis, key insights, and recommendations. Make the page visually appealing with clear sections, headings, and charts wherever needed. Include graphs to explain important patterns,*

and make sure everything is easy to understand. Also add a clean layout with proper spacing, colors, and readable text."